

Deep Deterministic Policy Gradient (DDPG)

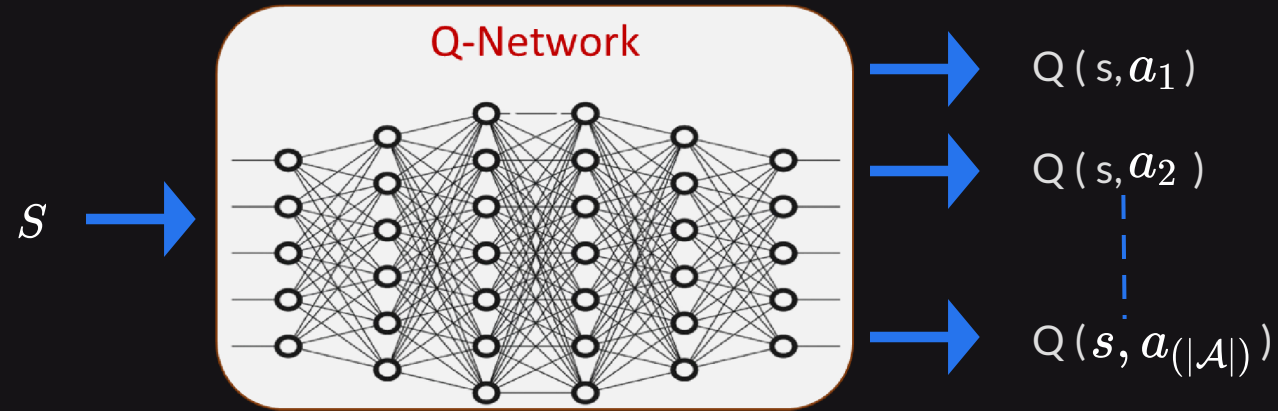
Instructor

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Optimal Action Selection With DQN

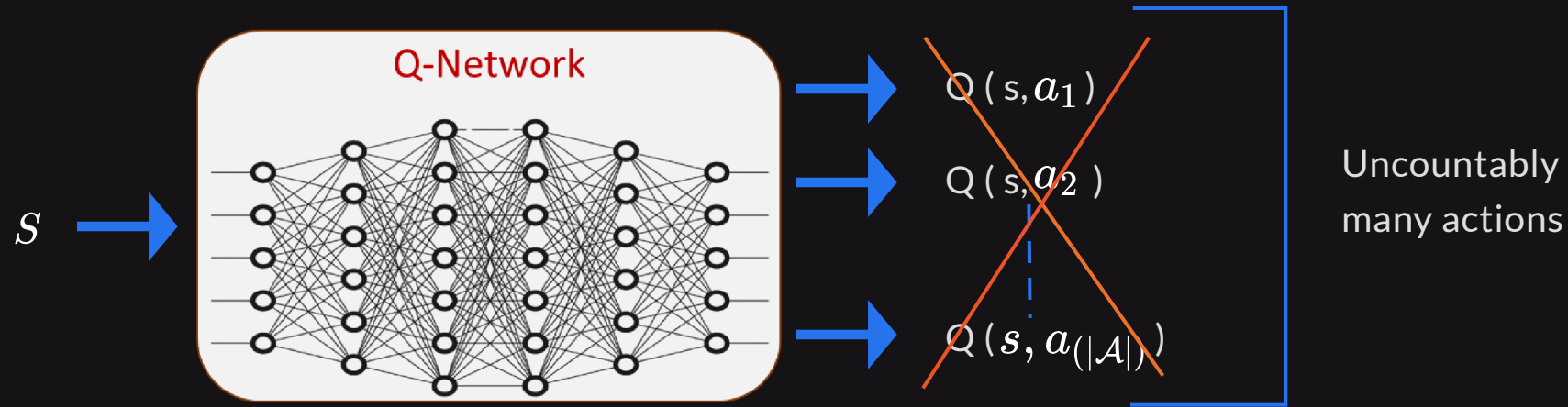


Policy:
$$\pi(s) = a^* = \arg \max_{a'} Q(s, a')$$

Also used in setting Q-targets at training time:

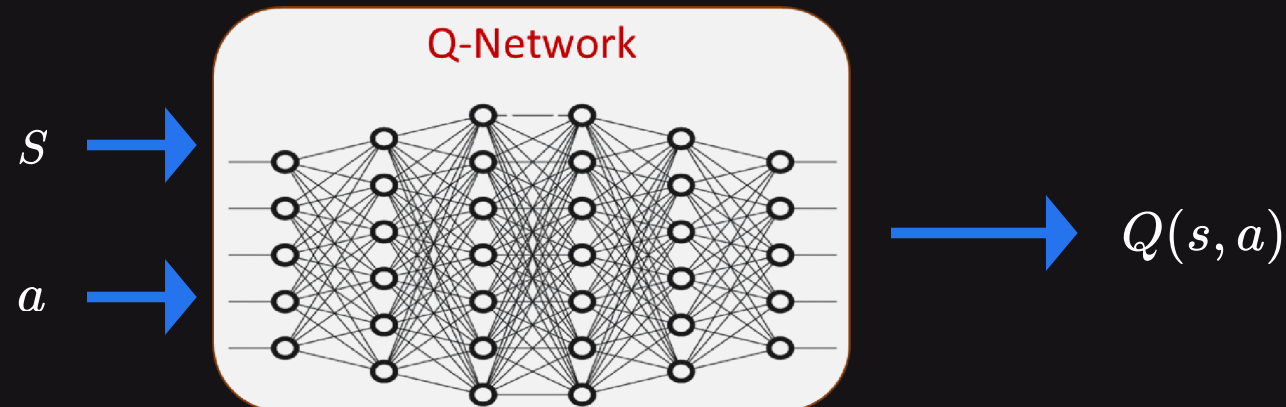
$$r + \gamma Q_{\phi}(s', a') = r + \gamma Q_{\phi}(s', \pi(s'))$$

DQN With Continuous Actions

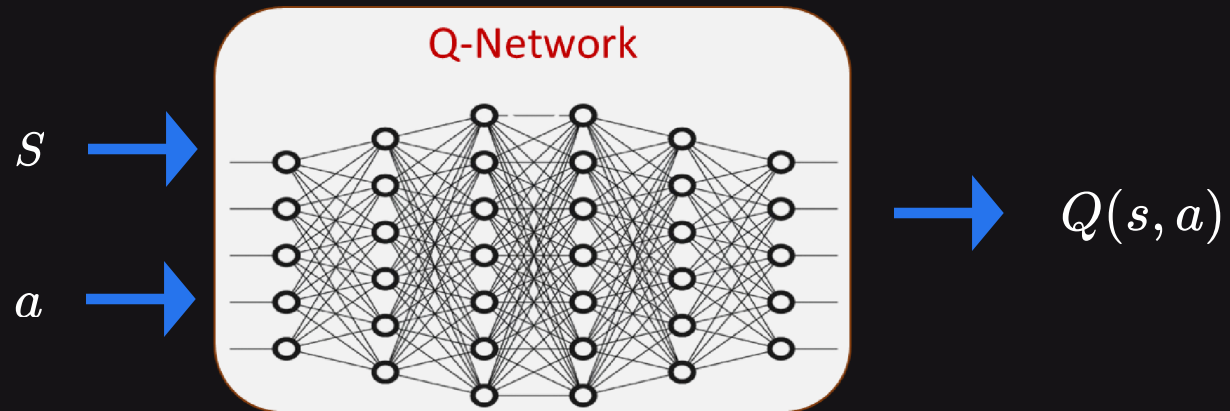


Cannot simply set $\pi(s) = a^* = \arg \max_{a'} Q(s, a')$

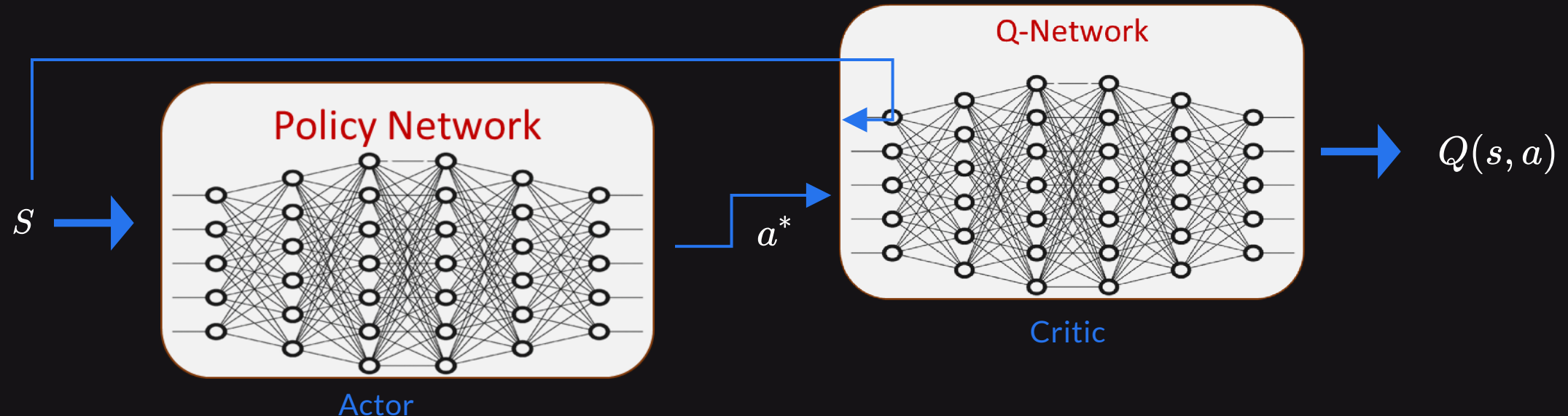
One possible approach:



DQN With Continuous Actions



- Will need to solve an optimization problem to find the argmax!
- Another approach:
 - Train a NN to directly produce the output of this maximization problem



Deep Deterministic Policy Gradient (DDPG)

- For iteration = 1, 2,

Rollouts:

Execute rollout under current policy (+ some noise for exploration)

Q-Update:

$$g \propto \nabla_{\phi} \sum_t (Q_{\phi}(s_t, u_t) - r_t - \gamma Q_{\phi}(s_{t+1}, u_{t+1}))^2$$

Policy Update:

Backpropagation through Q to compute gradient estimates for all t:

$$g \propto \sum_t \nabla_{\theta} Q_{\phi}(s_t, \pi_{\theta}(s_t, u_t))$$

Thank You
